ADAPTIVE DIGITAL SAT PRACTICE TEST

DSAT_Dec_2024_Int_1-W

Section 1: Reading & Writing — Module 1 (27 Qs) + Module 2 Easier (27 Qs) + Module 2 Harder (27 Qs)

Section 2: Math — Module 1 (22 Qs) + Module 2 Easier (22 Qs) + Module 2 Harder (22 Qs)

Total: 148 Questions

Section 1: Reading & Writing

Module 1 (27 Questions — All Students)

Question 1

Skill: Words in Context (Vocabulary) | Difficulty: Easy

The Pacific Islands are home to an extraordinary range of marine ecosystems, from the vast coral reefs surrounding Palau to the deep ocean trenches near Tonga. Much like the islands themselves, the variety of aquatic species across the Pacific is ______.

Which choice completes the text with the most logical and precise word or phrase?

A) staggering
B) mysterious
C) predictable
D) declining

Question 2

Skill: Words in Context (Vocabulary) | Difficulty: Easy

The dwarf planet 136472 Makemake was named after a Polynesian deity associated with creation, but given the enormous number of celestial objects that have been cataloged (more than 1.2 million to date), most are assigned only a numeric designation. Because any name given to a celestial object must be distinct from all others, naming each would be ______.

Which choice completes the text with the most logical and precise word or phrase?

A) unfeasible
B) controversial
C) redundant
D) habitual

Question 3

Skill: Words in Context (Vocabulary) | Difficulty: Medium

Commuters who firmly believe that the fare they must pay to ride the Bay Area Rapid Transit system, which serves cities throughout the San Francisco region, is currently too expensive are likely to be ______ proposals for raising the fare. Advocates for a higher fare are likely to find greater support if they instead direct their proposals toward a more receptive group of stakeholders.

Which choice completes the text with the most logical and precise word or phrase?

A) referenced in
B) informed of
C) resistant to
D) mollified by

Question 4

Skill: Words in Context (Vocabulary) | Difficulty: Medium

Though few scholars regard Mikhail Sholokhov's novel *Quiet Flows the Don*—which chronicles the lives of Cossack communities during the Russian Civil War—to be as artistically accomplished as his later work *The Fate of a Man*, some praise it despite the harm ______ wartime propaganda: literary historian Elena Volkov in *The Russian Literary Review* described the novel as "courageously unvarnished."

Which choice completes the text with the most logical and precise word or phrase?

A) conferred on
B) inflicted by
C) assigned to
D) hidden from

Question 5

Skill: Text Structure and Purpose | Difficulty: Easy

Researchers examining the ancient Roman settlement of Herculaneum in southern Italy recently uncovered a remarkably intact bakery known as a *pistrinum*. The site contains grinding stones, carbonized grain, and wall paintings. These artifacts provide scholars with a richer understanding of food production in Roman towns. For instance, the researchers discovered a stone oven that they believe was used to bake a flatbread made from emmer wheat.

Which choice best states the main purpose of the text?

A) To analyze the political hierarchy of ancient Rome
B) To describe a recent archaeological find and its significance
C) To compare agricultural practices across Mediterranean civilizations
D) To evaluate the nutritional quality of ancient Roman diets

Question 6

Skill: Text Structure and Purpose | Difficulty: Hard

Max Weber and other sociologists of industrialization rarely examine indigenous trade networks in Southeast Asia prior to the era of European colonialism, implicitly characterizing market-based commerce as a Western import to the region. Ariel Heryanto and other Southeast Asianist scholars have demonstrated, however, that in parts of the archipelago, sophisticated credit systems, the organized exchange of spices and textiles, and other hallmarks of commercial enterprise preceded colonial contact. One explanation for this oversight is that sociologists of industrialization tend to rely on quantitative trade records from colonial-era archives, which do not exist for much of precolonial Southeast Asia.

Which choice best describes the main purpose of the text?

A) To explain why sociologists and Southeast Asianist scholars interpret different commercial activities as evidence of market economies

B) To identify and account for a potentially misleading tendency in the work of sociologists of industrialization

C) To describe and assess a methodology used by Southeast Asianist scholars that sets their work apart from that of sociologists of industrialization

D) To outline a debate between sociologists of industrialization and Southeast Asianist scholars about how to analyze quantitative trade records from colonial archives

Question 7

Skill: Text Structure and Purpose | Difficulty: Hard

The following text is adapted from Charles Dickens's 1838 novel *Oliver Twist*. Mr. Bumble is a parish official in a small English town.

I am by no means certain that if the worthy citizens of the parish had discovered the full character of Mr. Bumble, they would not have been greatly astonished to learn that his conduct was rather less than entirely honorable. Simple judgments are comforting. It is far more convenient to settle in your mind that an official is beyond reproach, than to examine all the evidence that might compel you to revise that assessment.

Which choice best states the main purpose of the text?

A) To comment on a human inclination that may explain why the citizens of the parish regard Mr. Bumble as they do

B) To consider the conditions under which the negative reputation Mr. Bumble holds among the citizens could be changed

C) To argue that the citizens are largely justified in their critical view of Mr. Bumble's professional conduct

D) To invite the reader's sympathy for both Mr. Bumble and the citizens by presenting their contrasting perspectives

Question 8

Skill: Inferences | Difficulty: Medium

The following text is from Cristina García's 1992 novel *Dreaming in Cuban*. Felicia, a musician, is hosting visitors in the sitting room of her family's residence, and Herminia is her aunt. A *tertulia* is an informal cultural gathering.

It was nearly midnight when the last of the visitors rose to leave. Herminia had already appeared in the doorway three times to check whether the guests had departed. The back room had long been her personal domain, as she kept her sewing table and fabrics there. Now, each evening, it became Felicia's *tertulia*, as Herminia called it, and it was never restored to its proper arrangement before Herminia needed it the following afternoon.

Based on the text, what most likely motivates Herminia's behavior during Felicia's tertulia?

A) She is irritated because she requires help with household tasks, but Felicia is occupied with entertaining visitors.

B) She is eager for the gathering to conclude so that she can reclaim the space for her own purposes.

C) She is enthusiastic about joining the conversation with the visitors and hopes they will invite her to participate.

D) She finds the visitors uninteresting and is looking for a polite way to excuse herself from the tertulia.

Question 9

Skill: Command of Evidence (Textual) | Difficulty: Medium

Corporations engaged in timber harvesting count logging machinery among their capital investments. But timber harvesting is a highly resource-depleting activity, so as environmental regulations increasingly mandate sustainable forestry practices, demand for certain old-growth timber declines and the harvesting equipment will eventually, or even abruptly, become a financial burden as diminished timber revenues make it harder to justify the cost of maintaining such machinery.

What claim does the text make about the demand for old-growth timber?

A) Corporations involved in timber harvesting are unaffected by shifts in demand for old-growth timber.

B) The growing emphasis on environmental sustainability is a factor influencing demand for old-growth timber.

C) The demand for old-growth timber follows a consistent seasonal pattern.

D) Corporations involved in timber harvesting can profit from a reduction in demand for old-growth timber.

Question 10

Skill: Command of Evidence (Textual) | Difficulty: Medium

Why do certain gecko species produce chirping sounds while most lizards are silent? Biologists theorize that this distinction may be partly attributable to a structure in their larynx called the vocal fold. Chirping geckos, which are generally nocturnal, have a thick, flexible vocal fold that vibrates when air passes over it, producing sound. By contrast, diurnal lizards have a thin, rigid vocal fold, and the cartilage supporting it lacks the elasticity found in gecko species. These anatomical features allow geckos to produce their characteristic calls. The same traits may also explain why most other lizards remain silent.

According to the text, which trait do researchers believe may contribute to a gecko's ability to chirp?

A) The overall dimensions of the gecko's larynx

B) The cartilage connecting the vocal fold to the gecko's throat

C) The gecko's thick, flexible vocal fold

D) The relative body mass of the gecko compared to other lizards

Question 11

Skill: Command of Evidence (Quantitative) | Difficulty: Easy

[PLACEHOLDER: Table titled 'Total Areas of Five Alaskan Native Villages' with columns: Village | Area (square kilometers). Rows: Kivalina: 4.93 | Shaktoolik: 12.05 | Newtok: 8.31 | Quinhagak: 28.14 | Togiak: 17.82. Below the table is a passage: 'Alaskan Native villages are communities in the state of Alaska that have been designated as federally recognized tribal entities. The largest such village, Barrow, spans over 50 square kilometers along the Arctic coast. Most of the villages are considerably smaller. For example, the total area of Newtok is 8.31 square kilometers, and the total area of Kivalina is ______']

Which choice most effectively uses data from the table to complete the example?

A) 28.14 square kilometers.

B) 4.93 square kilometers.

C) 17.82 square kilometers.

D) 12.05 square kilometers.

Question 12

Skill: Command of Evidence (Textual) | Difficulty: Medium

The Great Gatsby is a 1925 novel by F. Scott Fitzgerald set on Long Island in the early 1920s. In the novel, Nick Carraway observes the lavish parties hosted by his mysterious neighbor, Jay Gatsby; Nick's cousin Daisy Buchanan is a frequent subject of Gatsby's attention. Gatsby envisions a reunion with Daisy:

Which quotation from *The Great Gatsby* best illustrates the claim?

A) "[Daisy] turned her head as there was a light dignified knocking at the front door, and Nick went out to the hall to greet Gatsby."

B) "Nick Carraway, settling into his rented cottage, gazed across the bay at the distant green light that marked the end of a dock."

C) "Though there was talk of newer, grander estates further along the shore, the established families still preferred the older mansions of East Egg."

D) "Already [Gatsby's] restless mind, racing ahead to the afternoon tea, the trembling handshake and the rekindled conversation, placed [Daisy] once again at his side."

Question 13

Skill: Inferences | Difficulty: Hard

Sound traveling through water encounters obstacles of varying density; by analyzing the reflected sound waves, marine animals can assess the size and distance of the obstacle. Research by Hiroshi Tanaka, Priya Mehta, and Carlos Vega using models of three dolphin rostrum shapes—short (high ratio of width to length), medium, and elongated (low ratio of width to length)—demonstrated that for low-frequency echoes, dolphins with medium-length rostrums would be more adept than short-snouted dolphins at distinguishing between reflected signals and ambient ocean noise. A second research team has therefore predicted that in turbid (murky) water conditions, medium-snouted dolphins will be more successful than short-snouted dolphins at detecting obstacles that produce low-frequency echoes.

Which finding, if true, would most directly support the second research team's prediction?

A) A study using obstacles that generated low-frequency echoes in turbid water found that the short-snouted river dolphin (*Platanista gangetica*) collided with obstacles more frequently than the medium-snouted bottlenose dolphin (*Tursiops truncatus*) did.

B) A study using obstacles that generated low-frequency echoes in turbid water found that the short-snouted harbor porpoise (*Phocoena phocoena*) navigated around obstacles as effectively as the medium-snouted common dolphin (*Delphinus delphis*).

C) A study using obstacles that generated low-frequency echoes in turbid water found that some specimens of the medium-snouted spinner dolphin (*Stenella longirostris*) collided with obstacles more often than other specimens of the same species.

D) A study using obstacles that generated low-frequency echoes in clear water found that the short-snouted river dolphin (*Platanista gangetica*), which has relatively small eyes, avoided most obstacles successfully.

Question 14

Skill: Command of Evidence (Quantitative) | Difficulty: Medium

[PLACEHOLDER: Table titled 'Population and Population Density of Southeast Asian Countries in 2020' with columns: Country | Density (inhabitants/km²) | Area (km²) | Estimated population. Rows: Indonesia: 151.0 | 1,904,569 | 273,500,000 / Thailand: 136.0 | 513,120 | 69,800,000 / Philippines: 368.0 | 300,000 | 109,600,000 / Cambodia: 95.0 | 181,035 | 16,720,000. Below: 'As the most populous region in the developing world, Southeast Asia was home to an estimated 668 million people in 2020. In a geography seminar, a student notes that countries with very large populations may be less densely populated than countries with much smaller populations, as can be seen by comparing ______']

Which choice most effectively uses data from the table to complete the example?

A) Indonesia, which has a low density, with the Philippines, which has a similar density despite both nations having different geographic sizes.

B) the geographic size of Thailand (513,120 square kilometers) with its relatively large population of 69,800,000 inhabitants.

C) Thailand, which has a large population of 69,800,000 inhabitants and a relatively low density of 136.0 inhabitants per square kilometer, with the Philippines, which has a much smaller area and a higher density.

D) the populations of both Indonesia and Thailand in 2020 with their populations in 2015.

Question 15

Skill: Inferences | Difficulty: Hard

Domestic cats show a notable preference for using their right paw, with roughly 70-80% favoring the right side during reaching tasks. Among studies of wild feline species, Dr. Sarah Jennings's 2004 study of captive lions and leopards reported more right-pawed individuals than left-pawed ones, but Dr. Marcus Webb and colleagues' 2011 study of wild African lions did not find a tendency toward right-pawedness. It is worth noting, however, that captive animals interact substantially more with human handlers than wild animals do, and therefore ______

Which choice most logically completes the text?

A) the apparent difference between the two studies' outcomes may be partly attributable to the 2004 study employing a different criterion to evaluate pawedness than the 2011 study used.

B) the right-pawedness observed in captive lions and leopards might be, in part, a consequence of prolonged exposure to human caretakers.

C) the number of subjects in the study of captive lions and leopards is probably insufficient to substantiate the conclusion that the species tends toward right-pawedness.

D) the number of subjects in the study of wild African lions is likely too limited to confirm that no right-paw preference exists in that population.

Question 16

Skill: Standard English Conventions (Punctuation/Possessives) | Difficulty: Medium

Employed in applications ranging from pharmaceuticals to food processing, stabilizers are compounds that prevent the separation of a suspension—a mixture in which solid particles are dispersed throughout a liquid. The stabilizers developed by ______ like carrageenan, enable these scientists to maintain uniform dispersions in commercial products such as syrups and paints.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) industrial chemists,

B) industrial chemist's,

C) industrial's chemists,

D) industrial chemists',

Question 17

Skill: Standard English Conventions (Punctuation) | Difficulty: Easy

Several early typewriters were operated by a rotating drum mechanism known as the Hansen Writing Ball. Hansen Writing Ball typewriters were widely used during the latter decades of the nineteenth ______ these distinctive instruments were gradually supplanted by more modern keyboard designs.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) century,

B) century that

C) century

D) century, but

Question 18

Skill: Standard English Conventions (Punctuation) | Difficulty: Easy

Ghanaian British sculptor El Anatsui uses reclaimed aluminum bottle caps to construct monumental tapestries of shimmering metal. His installation at the Venice Biennale honoring curator Okwui ______ is exhibited in galleries across four continents.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Enwezor:
B) Enwezor.
C) Enwezor
D) Enwezor—

Question 19

Skill: Standard English Conventions (Pronoun Usage) | Difficulty: Easy

The constitution of India, adopted in 1950, enshrines 6 fundamental rights across 22,000 words of text. According to constitutional analyst Pratap Bhanu Mehta, who studies the relationship between constitutional scope and governance, ______ ranked among the longest national constitutions worldwide.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) they're
B) their
C) it's
D) its

Question 20

Skill: Standard English Conventions (Sentence Boundaries) | Difficulty: Medium

Between late October and mid-November each year, debris shed by comet 55P/Tempel-Tuttle enters Earth's atmosphere at a velocity of 71 kilometers per second. Relatively few of these particles are likely to reach Earth's ______ since the meteoroids' interaction with the upper atmosphere causes the great majority to disintegrate in the thermosphere, producing the Leonid meteor shower.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) surface, however;
B) surface, however,
C) surface; however,
D) surface. However,

Question 21

Skill: Transitions | Difficulty: Medium

While most insects rely on the cover of darkness for safe dispersal, several species of brightly colored beetle—a group that includes the ladybug and the jewel beetle—routinely undertake flights during daylight hours. ______ with the sun overhead, the insects' vivid coloration deters visually oriented predators by functioning as an indicator of chemical defenses.

Which choice completes the text with the most logical transition?

A) At that time,
B) For instance,
C) Put differently,
D) Still,

Question 22

Skill: Transitions | Difficulty: Easy

Citizens of Tokyo, Japan, depend on its extensive subway network, the Tokyo Metro, for millions of daily commutes each year. ______ transit authorities work diligently to uphold the system's 285 stations to guarantee that each journey is as efficient as possible.

Which choice completes the text with the most logical transition?

A) For this reason,
B) Nevertheless,
C) For instance,
D) On the other hand,

Question 23

Skill: Transitions | Difficulty: Medium

Cel animation, employed by traditional artists to produce hand-drawn animated films, yields characters with fluid, organic movement. Admittedly, because each frame is drawn individually on transparent celluloid sheets, such hand-crafted animations require enormous labor; ______ digital compositing tools such as automated in-betweening are often necessary to achieve the volume of frames that modern audiences expect.

Which choice completes the text with the most logical transition?

A) as a result,
B) likewise,
C) to summarize,
D) on the other hand,

Question 24

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• Telegraph cables carried crucial electrical signals to connect distant cities before wireless communication.
• Before automation, telegraph stations were operated by trained telegraphists.
• Ada Spencer was the head telegraphist at the Whitby Junction Station in Yorkshire, England.
• She held this position from 1874 to 1896.
• Florence Nightingale Graham was the head telegraphist at the Cape Otway Station in Victoria, Australia.
• She held this position from 1901 to 1923.

The student wants to emphasize a difference between the two telegraphists. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) As the head telegraphist at Cape Otway Station, Florence Nightingale Graham helped connect distant Australian cities.

B) From 1874 to 1896, the communication networks of Yorkshire were maintained thanks to Ada Spencer.

C) Ada Spencer served as a head telegraphist in an earlier era than did Florence Nightingale Graham.

D) Ada Spencer and Florence Nightingale Graham both played important roles in maintaining long-distance communication networks in past centuries.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• If an asteroid passes close enough to a planet, gravitational forces can fracture the asteroid.
• In a 2021 paper, astrophysicists proposed that Jupiter was once orbited by a large asteroid they designated Theia.
• Their models suggested that Theia would likely have entered Jupiter's gravitational influence approximately 200 million years ago.
• At that proximity, Theia would have been shattered by tidal stresses.
• The astrophysicists theorized that the resulting fragments formed Jupiter's Trojan asteroids.

The student wants to recount the sequence of events proposed by the astrophysicists. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) According to the astrophysicists' models, fragments from Jupiter's Trojan asteroids would likely have caused Theia—a large asteroid orbiting Jupiter—to break apart approximately 200 million years ago.

B) In a 2021 paper, astrophysicists proposed that Jupiter was once orbited by a large asteroid that eventually fragmented.

C) Approximately 200 million years ago, a large asteroid (Theia) may have drifted close enough to Jupiter that gravitational forces shattered it; its fragments then formed the planet's Trojan asteroids.

D) If an asteroid (like those orbiting Jupiter) comes close enough to that planet, gravitational forces can cause the asteroid to fragment.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• *Freshwater Fish of East Africa* is an identification guide by ichthyologists David Seegers, Lothar Wischnath, and Erwin Schraml.
• It catalogs the forty-two cichlid species found in Lake Malawi.
• The electric blue hap is a large cichlid found in Lake Malawi.
• It is identifiable by its vivid blue coloring and its prominent forehead.
• The yellow lab cichlid is a small cichlid found in Lake Malawi.
• It is identifiable by its bright yellow body and its dark dorsal fin stripe.

Which choice most effectively uses information from the given sentences to emphasize a similarity between the two fish?

A) *Freshwater Fish of East Africa* is a carefully assembled reference by ichthyologists David Seegers, Lothar Wischnath, and Erwin Schraml.

B) Although they differ in several respects, the electric blue hap is larger than the yellow lab cichlid.

C) While each displays distinctive features of its own, both the electric blue hap and the yellow lab cichlid are found in Lake Malawi.

D) The yellow lab cichlid is a small fish identifiable by its bright yellow body and its dark dorsal fin stripe.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• Little is documented about the life of Laskarina Bouboulina (1771–1825).
• She was born on the island of Spetses, Greece, and earned renown as a naval commander and heroine of the Greek War of Independence.
• She achieved several remarkable military feats—some verified and some legendary.
• She has become an internationally recognized symbol of resistance thanks to her portrayal in dozens of novels, films, and television programs.
• In the 2007 television series *Captains of the Aegean*, actress Maria Kavoyianni portrays Bouboulina leading a daring naval blockade to liberate her homeland.

The student wants to emphasize the role of media in shaping Bouboulina's legacy. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Thanks to her depiction in dozens of media productions, Bouboulina was known locally as a successful naval commander and Greek independence heroine.

B) Bouboulina's exploits as a naval commander are featured in numerous media works, including the 2007 series *Captains of the Aegean*.

C) Multiple forms of media have depicted Bouboulina, the celebrated naval commander who became an internationally recognized symbol of resistance.

D) Though she was known locally during her lifetime, Bouboulina's subsequent portrayal in novels, films, and television has transformed her into an internationally recognized symbol of resistance.

Module 2 — Easier Path (27 Questions)

Approximately 70% Easy/Medium, 30% Hard

Question 1

Skill: Words in Context (Vocabulary) | Difficulty: Easy

Renowned for its exceptional collection of street murals, Philadelphia features works ranging from abstract geometric designs on warehouse walls to photorealistic portraits on residential buildings. The ______ variety of public artwork displayed throughout the city can thus captivate any visitor.

Which choice completes the text with the most logical and precise word or phrase?

A) remarkable
B) disputed
C) puzzling
D) indifferent

Question 2

Skill: Words in Context (Vocabulary) | Difficulty: Easy

In a 2019 review of documentaries exploring the heritage of Indigenous Australians, critics for the *Sydney Morning Herald* praised Warwick Thornton's 2017 film *We Don't Need a Map* as "electrifying" and Rachel Perkins's 2008 series *First Australians* as "profoundly moving." Admirers of the two productions hope that such ______ will draw wider viewership to these works.

Which choice completes the text with the most logical and precise word or phrase?

A) hesitation
B) recognition
C) caution
D) neutrality

Question 3

Skill: Words in Context (Vocabulary) | Difficulty: Medium

The skeletal remains of the specimen designated as Kennewick Man, discovered along the Columbia River in 1996, can help anthropologists not only ______ patterns in the migration of early peoples but also shed light on the Holocene epoch more broadly, uncovering significant details about the era in which Kennewick Man lived.

Which choice completes the text with the most logical and precise word or phrase?

A) obscure
B) trace
C) rank
D) perceive

Question 4

Skill: Text Structure and Purpose | Difficulty: Medium

Retailers are offering shoppers more options to personalize their purchases than ever before. Whether ordering custom phone cases, monogrammed luggage, or tailored suits, consumers can take part in shaping products to match their individual preferences. Companies benefit as well: research has demonstrated that buyers are prepared to spend more and accept longer delivery windows for a personalized item. Nevertheless, it can be challenging for manufacturers to provide customization while controlling expenses, as conventional assembly-line methods may not accommodate producing a unique item with each order.

Which choice best describes the overall structure of the text?

A) It discusses several recent technological breakthroughs in manufacturing and then proposes potential commercial uses for those breakthroughs.

B) It profiles a single company's recent achievement with new merchandise and then identifies multiple factors that may have driven that achievement.

C) It contrasts two opposing approaches to product marketing and then illustrates one of those approaches with examples.

D) It introduces a development in consumer goods and then explains how that development both advantages and poses a difficulty for companies.

Question 5

Skill: Text Structure and Purpose (Function) | Difficulty: Medium

The Riftwar Saga, first published in 1982, is a fantasy series by Raymond E. Feist, which spans over thirty interconnected novels. Some readers have criticized the uneven conclusions of *A Darkness at Sethanon* and other installments in the cycle, arguing that they neither resolve completely nor break off decisively. But other readers contend that the novels should not be assessed as independent works with conventional beginnings and endings but rather as chapters in a single sweeping epic, comparable to other multivolume sagas, such as Robert Jordan's *The Wheel of Time*.

Which choice best describes the function of the underlined portion in the text as a whole?

A) It presents a justification most readers use for why the Riftwar Saga should attain the same literary status as comparable works like *The Wheel of Time*.

B) It clarifies why many readers consider the Riftwar novels enjoyable in spite of shortcomings in their structure.

C) It claims that the unconventional format Feist employs for *A Darkness at Sethanon* makes it one of his most compelling installments.

D) It summarizes a favorable interpretation of a specific quality of the Riftwar novels.

Question 6

Skill: Text Structure and Purpose (Function) | Difficulty: Hard

Lateral gene transfer involves the movement of genetic material between organisms that are not in a direct ancestor-descendant relationship; vertical gene transfer, by contrast, is the passage of genetic material from parent to offspring. While lateral gene transfer is widespread among prokaryotes—single-celled organisms, including the bacteria *Escherichia coli* and *Staphylococcus aureus*—it has seldom been documented among eukaryotes (organisms with complex cells). Recent investigations, however, indicate that lateral gene transfer is more prevalent in eukaryotes than previously believed.

Which choice best states the function of the underlined sentence in the text as a whole?

A) It highlights a distinction between the mechanics of two categories of biological processes.

B) It contrasts the frequency with which a biological phenomenon has been identified in two groups of organisms.

C) It explains why a commonly held view of a biological mechanism is incorrect.

D) It argues that two biological phenomena are more closely related than they may initially seem.

Question 7

Skill: Cross-Text Connections | Difficulty: Hard

Text 1

In independent experiments, Dr. Kenji Matsuda and Dr. Ruth Adeyemi and colleagues investigated whether trees share carbohydrates with one another through a shared mycorrhizal network (SMN)—an underground web of fungal threads in the soil. Matsuda and Adeyemi excluded all transfer routes other than the SMN by installing barriers to isolate the trees' root systems while permitting mycorrhizal threads to pass through—a critical precaution that the study by Dr. Liu and colleagues did not implement.

Text 2

Matsuda and Adeyemi took the essential step of separating the trees' root systems (thereby eliminating root-to-root exchange). However, any barrier employed must permit the fine fungal hyphae of an SMN to penetrate, and this permeability would also allow dissolved sugars to seep through. Consequently, the researchers' experimental design cannot guarantee that any carbohydrate transfer detected is attributable to an SMN rather than to some alternative pathway.

Based on the texts, the author of Text 1 and the author of Text 2 would most likely agree with which statement?

A) The barriers used in Matsuda and Adeyemi's study successfully prevented root-to-root carbohydrate exchange.

B) A barrier impervious to both roots and fungal threads is required to assess carbohydrate transfer via an SMN.

C) Preventing root-to-root carbohydrate exchange between trees is adequate to ensure that any observed transfer must involve an SMN.

D) Liu and colleagues' study did not produce persuasive evidence of carbohydrate transfer between individual trees.

Question 8

Skill: Command of Evidence (Textual) | Difficulty: Medium

Optimal foraging theory (OFT) proposes that animals' feeding strategies reflect cost-benefit calculations that differ by species and with shifting environmental conditions. One such condition is ambient temperature, which Dr. Thomas Reed and colleagues found to be negatively correlated with foraging by alpine marmots but which Dr. Hana Petrova and colleagues found to be positively correlated with foraging by desert iguanas. This discrepancy is explicable in terms of OFT: the iguanas' dependence on external warmth for metabolic activity means that higher temperatures facilitate foraging for them more than for the marmots.

Which choice best describes the finding made by Petrova and colleagues, as presented in the text?

A) Higher ambient temperature benefits desert iguanas' foraging more than it benefits alpine marmots' foraging.

B) As ambient temperature rises, desert iguanas increase their foraging activity.

C) During periods of elevated temperature, desert iguanas increase their reliance on thermoregulatory behavior.

D) As ambient temperature rises, alpine marmots reduce their foraging activity.

Question 9

Skill: Command of Evidence (Quantitative) | Difficulty: Easy

Entomologist Dr. Lydia Franklin and colleagues conducted a study examining how moth wing pattern and atmospheric pressure relate to moth flight behavior, which was made possible by observations recorded by volunteers and amateur naturalists in the region. Analyzing over four years of data, the researchers determined that brown moths were spotted near artificial lights more frequently than any other color variety, and that moths were observed to fly more often on low-pressure evenings than on high-pressure evenings.

According to the text, which factors appeared to be connected to moth flight behavior in Franklin and colleagues' study?

A) Atmospheric pressure but not moth wing pattern

B) Moth wing pattern but not atmospheric pressure

C) Neither moth wing pattern nor atmospheric pressure

D) Both moth wing pattern and atmospheric pressure

Question 10

Skill: Command of Evidence (Textual) | Difficulty: Hard

The primate species *Cebus capucinus* (the white-faced capuchin) shares habitat in Costa Rican forests with *Alouatta palliata* (the mantled howler monkey), which produces a resonant vocalization when it senses a threat. Primatologist Dr. Lucia Vargas and colleagues recorded *A. palliata* alarm vocalizations and played them in the vicinity of wild *C. capucinus*. Observing that the capuchins frequently ascended to higher branches or ceased movement upon hearing the calls, the researchers concluded that *C. capucinus* interprets *A. palliata* alarm vocalizations as warning signals.

Which finding, if true, would most directly support Vargas and colleagues' conclusion?

A) Other primate species also tended to ascend or stop moving when Vargas and colleagues played *A. palliata* alarm vocalizations.

B) Vargas and colleagues played alarm vocalizations from multiple *A. palliata* individuals and found no meaningful variation in the reactions of *C. capucinus*.

C) In certain instances, *C. capucinus* climbed higher or froze when Vargas and colleagues approached but before any sounds were played.

D) When Vargas and colleagues played recordings of neutral ambient forest sounds near *C. capucinus*, the monkeys exhibited no observable response.

Question 11

Skill: Command of Evidence (Quantitative) | Difficulty: Medium

[PLACEHOLDER: Table titled 'Studies of Gray Wolf Population Density' with columns: Study authors | Location | Methods | Minimum density (wolves per 100 sq km) | Maximum density (wolves per 100 sq km) | Density range. Rows: Dr. James Harper et al. | Montana (US) | GPS collar tracking | 2.10 | 3.90 | 1.80 / Dr. Omar Haddad et al. | British Columbia (Canada) | scat-analysis dogs | 1.95 | 4.60 | 2.65 / Dr. Yuki Sato et al. | Hokkaido (Japan) | motion-sensor cameras | 4.72 | 8.60 | 3.88 / Dr. Elena Marchetti et al. | Abruzzo (Italy) | howl-response surveys, GPS tracking | 1.30 | 1.55 | 0.25. Below: 'Researchers have employed various techniques to estimate the population density of gray wolves (Canis lupus). A student claims that the use of scat-analysis dogs yields the most precise results, with the smallest difference between minimum and maximum densities.']

Which choice best describes data from the table that weaken the student's claim?

A) Dr. Yuki Sato et al. found a density range of 3.88 individuals per 100 square kilometers despite their use of motion-sensor cameras.

B) Dr. Elena Marchetti et al. found a density range that was substantially smaller than that found by Dr. Yuki Sato et al.

C) Dr. Omar Haddad et al. found a density range greater than that found by some studies that employed other techniques.

D) Dr. Omar Haddad et al. found a maximum density of 4.60 individuals per 100 square kilometers, which differed from that found by some studies using other techniques.

Question 12

Skill: Command of Evidence (Textual) | Difficulty: Hard

The language complexity hypothesis (LCH) asserts that the proportion of non-native speakers in a language community (its exotericity) and the grammatical complexity of that language are negatively correlated, a relationship the LCH attributes to the erosion of intricate grammatical structures as more non-native speakers adopt the language but struggle with those structures. Examining two features that serve as positive markers of grammatical complexity—agglutination (when multiple meaningful units combine into single words) and tonal distinction (languages' use of pitch to differentiate meaning)—Dr. Yuri Petrov and colleagues performed a statistical analysis across more than 1,500 languages and assert that the results are inconsistent with the LCH.

Which finding, if true, would most directly support Petrov and colleagues' assertion?

A) Petrov and colleagues' analysis revealed a slightly positive correlation between grammatical complexity and agglutination.

B) Petrov and colleagues' analysis revealed a slightly negative correlation between grammatical complexity and exotericity.

C) Petrov and colleagues' analysis revealed a slightly positive correlation between agglutination and exotericity and between tonal distinction and exotericity.

D) Petrov and colleagues' analysis revealed a slightly negative correlation between grammatical complexity and agglutination and between grammatical complexity and tonal distinction.

Question 13

Skill: Command of Evidence (Quantitative) | Difficulty: Hard

[PLACEHOLDER: Bar chart with title 'Correlation Between Model Predictions and Participant Ratings by Painting Style.' X-axis: Painting style (Landscape, Still life). Y-axis: Correlation (0 to 0.3). Three bars per style for participants S8 (dark gray), S12 (light gray), S10 (black). For Landscape: S8 ≈ 0.27, S12 ≈ 0.28, S10 ≈ 0.23. For Still life: S12 ≈ 0.22, S8 ≈ 0.17, S10 ≈ 0.23. Below: passage about neuroscientist Dr. Akira Nakamura developing a computational model to predict art enjoyment on a 1-4 scale, recruiting participants, and calculating correlation. 'Assuming participant S12 gave equal ratings to the landscape and still-life paintings, the data in the graph suggest that the model predicted that ______']

Which choice most effectively uses data from the graph to complete the statement?

A) Participant S12's ratings for landscape and still-life paintings would diverge from one another.

B) Participant S12 would find landscape paintings more aesthetically pleasing than still-life paintings.

C) Participant S12's ratings for landscape and still-life paintings would be approximately equal.

D) Participant S12 would find landscape paintings less aesthetically pleasing than still-life paintings.

Question 14

Skill: Inferences | Difficulty: Medium

Norse sagas (tales about Scandinavian heroes and gods) derive from numerous often inconsistent oral traditions, such as the *Prose Edda*, compiled around 1220, and *Gesta Danorum* from approximately 1200. Snorri Sturluson's 13th-century *Prose Edda* was an effort to systematize these stories into a unified mythology. Many of Sturluson's sources draw from older *skaldic* poetry, composed in the 900s. While neither the *skaldic* poems nor any earlier sources mention Thor's iconic hammer Mjölnir as returning to his hand after being thrown, the *Prose Edda* does, suggesting that ______

Which choice most logically completes the text?

A) when an oral tradition conflicted with the version in the *Prose Edda*, Sturluson chose to include the older account.

B) The *Prose Edda* is more historically reliable than the *skaldic* poetry, because *Gesta Danorum* had not yet been compiled when the poems were composed.

C) The *skaldic* poets' depictions of Norse myths are more consistent overall with the *Gesta Danorum* than with the *Prose Edda*.

D) Sturluson drew on a tradition that the *skaldic* poets did not have access to when composing their works.

Question 15

Skill: Inferences | Difficulty: Hard

Extended exposure to artificial light at night (ALAN) can alter the behavior of wildlife, as Dr. Rebecca Torres and Dr. Hamish Campbell documented in a 2016 study of urban sparrows. Researchers conducted a meta-analysis of studies examining how ALAN affects animals and found that, for every study reviewed, measurable behavioral or physiological changes were evident between exposed and unexposed groups. Although, on average, studies of mammals showed larger effect sizes than studies of birds did, for every taxonomic class examined, there were individual studies showing effects well above the average for mammals. Therefore, the results of the meta-analysis suggest that ______

Which choice most logically completes the text?

A) the studies in the meta-analysis examining mammals were more likely than those examining birds to specify whether the observed effects were harmful.

B) the effect size documented by Torres and Campbell was probably larger than the average effect size for studies of sparrows included in the meta-analysis.

C) certain studies of birds found greater effects of exposure to ALAN than certain studies of mammals did.

D) the effects that studies attribute to ALAN exposure are likely to be more pronounced for birds than for mammals.

Question 16

Skill: Standard English Conventions (Punctuation) | Difficulty: Medium

The word "uvular" comes from Latin and means "relating to the uvula," an appropriate label for the category of consonants—uvular consonants—produced by raising the back of the tongue toward the uvula at the rear of the mouth. In most Romance languages, including French, these consonants are ______ in several Caucasian languages, however, five or more such consonants ("q," "χ," "■," among others) are regularly used.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) uncommon,
B) uncommon
C) uncommon;
D) uncommon and

Question 17

Skill: Standard English Conventions (Punctuation) | Difficulty: Medium

Together with soil moisture levels and nutrient availability, sunlight exposure influences the growth rate of ______ as sunlight exposure increases, so does the rate at which carbon dioxide and water are transformed into sugars and oxygen through this process.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) germination
B) germination:
C) germination and
D) germination,

Question 18

Skill: Standard English Conventions (Verb Forms) | Difficulty: Medium

The lichen *Xanthoria parietina*, commonly known as the golden shield lichen, is recognized for establishing mutualistic associations with certain birch tree species; in many instances, such symbiotic pairings, in which fungal filaments intertwine with the birch's bark, ______ expansive colonies, which can facilitate the exchange of moisture between organisms.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to develop
B) having developed
C) develop
D) developing

Question 19

Skill: Standard English Conventions (Punctuation/Appositives) | Difficulty: Medium

Although he would later collaborate with numerous acclaimed composers, including Arvo Pärt on his album *Tabula Rasa Sessions*, Estonian ______ may be most celebrated for his tenure with the Tallinn Chamber Orchestra, where he helped shape the sound of contemporary classical music.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) conductor, Tõnu Kaljuste
B) conductor Tõnu Kaljuste,
C) conductor Tõnu Kaljuste
D) conductor, Tõnu Kaljuste,

Question 20

Skill: Standard English Conventions (Punctuation) | Difficulty: Medium

Dr. Miriam Goldstein's extended essay on "bioluminescence," a term describing the emission of light by living organisms, is featured in the anthology *Luminous Seas: A Guide to Marine Biology*. The anthology was not composed solely by ______ other marine scientists, such as Dr. James Whitfield and Dr. Ana Reyes, also contributed chapters.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Goldstein, however,
B) Goldstein; however,
C) Goldstein. However,
D) Goldstein, however;

Question 21

Skill: Standard English Conventions (Punctuation) | Difficulty: Easy

The International Monetary Fund (IMF) publishes comparative cost-of-living statistics for its member nations. According to these figures, in March 2022, a standard set of consumer goods valued at 100 US dollars (USD) in the United States would have cost 72 USD in fellow member ______

Which choice completes the text so that it conforms to the conventions of Standard English?

A) nation Poland.
B) nation: Poland.
C) nation, Poland.
D) nation—Poland.

Question 22

Skill: Transitions | Difficulty: Medium

In her sculpture *Perpetual Motion*, Maya Lin seeks to represent the flow of water in five-dimensional space. Naturally, Lin's piece cannot literally depict a fifth dimension; human perception is confined to three spatial dimensions. ______ by capturing the water's movement through layered transparent panels viewed from multiple angles simultaneously, Lin creates a compelling impression of what such a higher-dimensional reality might resemble.

Which choice completes the text with the most logical transition?

A) For example,

B) That said,

C) In other words,

D) Furthermore,

Question 23

Skill: Transitions | Difficulty: Easy

In 2018, cartographer Dr. Elaine Nakamura recalculated the geographic center of Australia, which had originally been determined in 1988 by the Royal Geographic Society. The Society, Nakamura argued, had overlooked critical variables, including offshore island territories. ______ the Society's established center—a point 35 kilometers south of Alice Springs—was imprecise.

Which choice completes the text with the most logical transition?

A) Additionally,

B) In contrast,

C) Therefore,

D) Nevertheless,

Question 24

Skill: Transitions | Difficulty: Easy

On March 12, 1947, test pilot Chuck Yeager was honored with a civic reception in Hamlin, West Virginia, in celebration of his record-breaking supersonic flight. Of the 318 civic receptions held between 1920 and 2000, a significant number recognized achievements in aviation. ______ Yeager's ceremony was just one of 22 honoring aviators.

Which choice completes the text with the most logical transition?

A) Moreover,

B) Consequently,

C) Indeed,

D) Nonetheless,

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• The Bauhaus was a pioneering German art and design school established in 1919 by Walter Gropius.
• Gropius directed the school from 1919 to 1928.
• Hannes Meyer served as director from 1928 to 1930.
• Ludwig Mies van der Rohe directed it from 1930 to 1933.
• Josef Albers was a painter and educator who began studying at the Bauhaus in 1920.
• Anni Albers was a textile artist and designer who began studying at the Bauhaus in 1922.

The student wants to highlight the chronological order in which the two students enrolled. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Walter Gropius founded the Bauhaus in 1919 and led it until 1928, when Hannes Meyer assumed the directorship.

B) Josef Albers and Anni Albers were both Bauhaus students, but Josef, who enrolled in 1920, preceded Anni by two years.

C) Established in 1919, the Bauhaus was a German art and design institution whose alumni include Josef Albers and Anni Albers.

D) The Bauhaus was founded in 1919, and Anni Albers began her studies there in 1922.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:
• In general, a material will expand when heated.
• The expansion of a material due to heat is known as thermal expansion.
• A 2020 study led by Dr. Wei Zhang and Dr. Carlos Rivera tested the thermal expansion of various metal alloys.
• When a 2-millimeter-thick sample of titanium alloy was heated, its average length increased by 0.9 micrometers.
• When a 5-millimeter-thick sample of copper-nickel alloy was heated, its average length increased by 4.2 micrometers.

The student wants to contrast the two samples. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The copper-nickel alloy sample tested in the 2020 study was thicker than the titanium alloy sample.

B) Heating a material will typically cause it to increase in size, a phenomenon called thermal expansion.

C) When the alloys were heated as part of the 2020 study, the length of both samples grew.

D) In 2020, researchers examined the thermal expansion of various metal alloys, including samples of titanium alloy and copper-nickel alloy.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

• Mineral abundance is a comparative measure of the quantity of a mineral in a particular geological formation.
• Abundance can be expressed as a mass fraction.
• A mass fraction is the ratio of a mineral's total mass to the combined mass of all minerals in a formation.
• The mass fraction of quartz (SiO_2) in Earth's continental crust is 120,000 parts per million (ppm).
• The mass fraction of zircon ($ZrSiO_4$) in Earth's continental crust is 165 ppm.

The student wants to specify the total mass of Earth's quartz relative to that of all minerals. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The mass fraction of quartz in Earth's continental crust is 120,000 ppm.

B) Quartz's mass relative to the combined mass of all minerals in Earth's crust is 165 ppm.

C) Quartz's abundance can be expressed by a mass fraction, a measure of the mineral's mass relative to that of all minerals in Earth's continental crust.

D) Quartz's abundance in Earth's crust exceeds that of zircon, as reflected by quartz's higher mass fraction.

Module 2 — Harder Path (27 Questions)

Approximately 30% Easy/Medium, 70% Hard

Question 1

Skill: Words in Context (Vocabulary) | Difficulty: Medium

The Andean highlands harbor an impressive diversity of tuber crops, from the widely cultivated potato grown in terraced fields at 4,000 meters to the lesser-known oca harvested by indigenous communities in Bolivia. The ______ range of cultivated root vegetables across the region reflects centuries of agricultural adaptation.

Which choice completes the text with the most logical and precise word or phrase?

A) expansive
B) contentious
C) bewildering
D) diminishing

Question 2

Skill: Words in Context (Vocabulary) | Difficulty: Medium

In a 2020 analysis of graphic novels portraying immigrant experiences, reviewers for *The Guardian* lauded Shaun Tan's 2006 work *The Arrival* as "breathtakingly inventive" and Marjane Satrapi's 2000 memoir *Persepolis* as "bracingly honest." Supporters of these creators anticipate that such ______ will introduce broader readerships to the genre.

Which choice completes the text with the most logical and precise word or phrase?

A) apprehension
B) endorsement
C) prudence
D) detachment

Question 3

Skill: Words in Context (Vocabulary) | Difficulty: Hard

The archaeological site known as Göbekli Tepe, unearthed in southeastern Turkey in 1994, can help prehistorians not only ______ transitions in the development of early agrarian societies but also illuminate the Neolithic period more broadly, revealing crucial information about the epoch in which the site's builders lived.

Which choice completes the text with the most logical and precise word or phrase?

A) manufacture
B) reconstruct
C) minimize
D) authenticate

Question 4

Skill: Text Structure and Purpose | Difficulty: Hard

The following text is adapted from Willa Cather's 1918 novel *My Ántonia*. Jim Burden is reflecting on his childhood in rural Nebraska.

I do not think anyone could have persuaded me, in those early years, that life beyond the prairie held any advantage over what I already possessed. The familiar is always simpler to cherish. It requires no argument, no comparison with alternatives. One needs only to look at the horizon, unchanging and absolute, to feel that no other view could possibly satisfy.

Which choice best states the main purpose of the text?

A) To reflect on a psychological tendency that may account for why Jim feels attached to the prairie landscape

B) To describe the specific geographic features of the Nebraska prairie that Jim finds appealing

C) To argue that Jim's preference for rural life is objectively superior to urban alternatives

D) To present contrasting perspectives on the value of familiarity versus novelty in shaping identity

Question 5

Skill: Text Structure and Purpose (Function) | Difficulty: Hard

Epigenetic modification involves changes to gene expression without alterations to the underlying DNA sequence; genetic mutation, conversely, involves direct changes to the nucleotide sequence itself. While epigenetic modification is well documented in multicellular organisms—complex life forms such as the plant *Arabidopsis thaliana* and the nematode *Caenorhabditis elegans*—it was long thought to be absent in unicellular organisms (single-celled life). Emerging research, however, suggests that epigenetic modification occurs in single-celled organisms more frequently than scientists had assumed.

Which choice best states the function of the underlined sentence in the text as a whole?

A) It draws a contrast between the prevalence of a biological process in two categories of organisms.

B) It explains why a previously accepted model of a molecular mechanism has been abandoned.

C) It identifies a methodological limitation that has hindered research into unicellular organisms.

D) It provides evidence that two biological processes are functionally equivalent.

Question 6

Skill: Cross-Text Connections | Difficulty: Hard

Text 1

In parallel investigations, Dr. Samuel Okafor and Dr. Ingrid Halvorsen and colleagues tested whether coral colonies transfer defensive compounds to neighboring colonies via a shared chemical signaling network (CSN)—a dissolved chemical pathway in surrounding seawater. Okafor and Halvorsen controlled for all dispersal mechanisms other than the CSN by using impermeable enclosures to isolate individual colonies while allowing dissolved chemicals to diffuse through—a methodological step that Dr. Park and colleagues' study omitted.

Text 2

Okafor and Halvorsen appropriately controlled for direct tissue-to-tissue contact between coral colonies (thereby ruling out contact transfer). However, the enclosures used must permit dissolved chemical signals to pass through, and this permeability would also allow waterborne bacteria to traverse the barrier. Thus, the researchers' experimental design cannot establish that any defensive compound transfer detected is attributable solely to a CSN rather than to microbial vectors.

Based on the texts, the author of Text 1 and the author of Text 2 would most likely agree with which statement?

A) The enclosures used in Okafor and Halvorsen's study effectively prevented tissue-to-tissue compound transfer.

B) An enclosure that blocks both tissue contact and dissolved chemicals is necessary to evaluate compound transfer via a CSN.

C) Eliminating tissue-to-tissue contact between coral colonies is sufficient to ensure that any detected compound transfer must involve a CSN.

D) Park and colleagues' study failed to detect meaningful evidence of defensive compound transfer between coral colonies.

Question 7

Skill: Command of Evidence (Textual) | Difficulty: Hard

The competitive exclusion principle (CEP) holds that two species competing for identical resources in the same habitat cannot coexist indefinitely—one will inevitably outcompete the other. Dr. Fatima Al-Rashid and colleagues, however, observed two closely related species of dung beetle (*Onthophagus taurus* and *Onthophagus gazella*) coexisting in the same Australian pasture for over a decade. Al-Rashid's team hypothesized that subtle differences in diel activity patterns—*O. taurus* being predominantly nocturnal and *O. gazella* predominantly diurnal—allow niche partitioning sufficient to avoid competitive exclusion.

Which finding, if true, would most directly support Al-Rashid and colleagues' hypothesis?

A) When both species were confined to an enclosure with abundant dung available only during daylight hours, *O. gazella* significantly outperformed *O. taurus* in reproduction.

B) In pastures where artificial lighting eliminated the distinction between day and night, one of the two species declined markedly within three years.

C) Genetic analysis revealed that *O. taurus* and *O. gazella* diverged from a common ancestor approximately 2 million years ago.

D) Both species were found to prefer dung from cattle over dung from sheep when given a choice.

Question 8

Skill: Command of Evidence (Textual) | Difficulty: Hard

Cognitive load theory (CLT) proposes that the processing demands of a task influence performance outcomes, with higher demands generally leading to more errors. Dr. Anika Patel and colleagues found that surgeons made significantly more technical errors during high-complexity laparoscopic procedures than during low-complexity ones. However, Dr. Tomoko Ishikawa and colleagues found that experienced air traffic controllers showed *no* significant increase in errors when managing high-traffic versus low-traffic airspace. This apparent contradiction is reconcilable within CLT: extensive domain-specific training can automate certain cognitive processes, effectively reducing the experienced cognitive load even as objective task complexity rises.

Which choice best describes the finding made by Ishikawa and colleagues, as presented in the text?

A) Increased task complexity is more detrimental to surgeon performance than to air traffic controller performance.

B) As the volume of air traffic increased, experienced controllers maintained a stable error rate.

C) During high-traffic periods, experienced controllers compensated by increasing their reliance on automated systems.

D) As task complexity rose, surgeons reduced their error rate through domain-specific training.

Question 9

Skill: Command of Evidence (Quantitative) | Difficulty: Hard

[PLACEHOLDER: Bar chart titled 'Correlation Between Climate Model Predictions and Actual Crop Yields by Crop Type.' X-axis: Crop type (Grain, Legume). Y-axis: Correlation (0 to 0.35). Three grouped bars per crop type for regions X (dark gray), Y (light gray), Z (black). For Grain: X ≈ 0.30, Y ≈ 0.32, Z ≈ 0.25. For Legume: X ≈ 0.18, Y ≈ 0.24, Z ≈ 0.26. Below: agricultural scientist passage about developing a climate model to predict crop yields, calculating correlation. 'Assuming Region X devoted equal acreage to grain and legume cultivation, the data suggest that the model predicted that ______']

Which choice most effectively uses data from the graph to complete the statement?

A) Region X's output for grain and legume crops would differ from one another.

B) Region X would produce more grain crops than legume crops.

C) Region X's output for grain and legume crops would be approximately equal.

D) Region X would produce fewer grain crops than legume crops.

Question 10

Skill: Inferences | Difficulty: Hard

Medieval European illuminated manuscripts (texts adorned with decorative illustrations) draw from numerous, often conflicting artistic traditions, such as the *Book of Kells*, created around 800 CE, and the *Très Riches Heures du Duc de Berry* from approximately 1416. Jean Colombe's 15th-century completion of the *Très Riches Heures* was an attempt to harmonize earlier artistic contributions into a unified visual program. Many of Colombe's stylistic choices draw from the Limbourg Brothers' original illustrations, painted in the 1410s. While neither the Limbourg Brothers' sections nor any earlier illuminated works depict the patron duke with a halo of gold leaf, Colombe's additions do, suggesting that ______

Which choice most logically completes the text?

A) when an artistic tradition in illuminated manuscripts contradicted the style of the *Très Riches Heures*, Colombe preferred to follow the Limbourg Brothers' approach.

B) The *Très Riches Heures* is more artistically significant than the *Book of Kells*, because the *Book of Kells* had not been widely circulated when Colombe was working.

C) The Limbourg Brothers' illustrations are more consistent overall with the *Book of Kells* than with Colombe's later additions to the *Très Riches Heures*.

D) Colombe drew on an iconographic convention that the Limbourg Brothers had not employed when creating their portion of the manuscript.

Question 11

Skill: Inferences | Difficulty: Hard

Chronic exposure to elevated ambient noise (sounds from anthropogenic sources such as construction or highway traffic) can influence marine organisms, as Dr. Rachel Simmons and Dr. Gregor Wolff demonstrated in a 2018 study of reef-dwelling damselfish. Scientists conducted a meta-analysis of research examining how such noise affects aquatic animals and found that, for every study reviewed, relevant behavioral or physiological traits differed measurably between noise-exposed and control groups. Although, on average, studies of invertebrates showed larger differences than studies of fish did, for every taxonomic class analyzed, there were individual studies showing differences well above the invertebrate average. Therefore, the meta-analysis results suggest that ______

Which choice most logically completes the text?

A) the studies in the meta-analysis examining invertebrates were more likely than those examining fish to determine whether the documented effects were detrimental.

B) the difference documented by Simmons and Wolff was probably larger than the average difference for damselfish studies included in the meta-analysis.

C) some studies of fish found greater effects of noise exposure than some studies of invertebrates did.

D) the effects that studies attribute to ambient noise exposure are likely to be more pronounced for invertebrates than for fish.

Question 12

Skill: Command of Evidence (Textual) | Difficulty: Hard

The insect species *Formica rufa* (the red wood ant) inhabits forested regions of northern Europe alongside *Lasius niger* (the common black ant), which releases a pungent chemical alarm signal when it encounters a predator. Entomologist Dr. Petra Engström and colleagues collected *L. niger* alarm secretions and dispersed them near colonies of wild *F. rufa*. Finding that the red wood ants frequently retreated into their mound or adopted defensive postures upon detecting the secretions, the researchers concluded that *F. rufa* recognizes *L. niger* alarm chemicals as danger cues.

Which finding, if true, would most directly support Engström and colleagues' conclusion?

A) Other ant species besides *F. rufa* also displayed defensive behaviors when Engström and colleagues dispersed *L. niger* alarm secretions.

B) Engström and colleagues used alarm secretions from multiple *L. niger* colonies and observed no significant variation in *F. rufa* responses.

C) In some cases, *F. rufa* retreated into their mound when Engström and colleagues approached but before any chemical was released.

D) When Engström and colleagues dispersed a neutral, odorless control substance near *F. rufa* colonies, the ants showed no behavioral change.

Question 13

Skill: Standard English Conventions (Punctuation) | Difficulty: Medium

The term "pharyngeal" derives from Greek and means "relating to the pharynx," a fitting descriptor for the class of speech sounds—pharyngeal consonants—articulated by constricting the pharynx at the back of the throat. In most European languages, including German, these consonants are ______ in several Semitic languages, however, three or more such sounds are regularly produced.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) absent,

B) absent

C) absent;

D) absent and

Question 14

Skill: Standard English Conventions (Punctuation) | Difficulty: Hard

In addition to atmospheric humidity and wind speed, barometric pressure affects the evaporation rate of ______ as barometric pressure decreases, so does the boiling point of the liquid, accelerating the transition from liquid to vapor.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) water

B) water:

C) water and

D) water,

Question 15

Skill: Standard English Conventions (Verb Forms) | Difficulty: Hard

The bacterium *Rhizobium leguminosarum*, widely recognized as a nitrogen-fixing microbe, is known to establish mutualistic associations with certain legume species; in many cases, such symbiotic relationships, wherein bacterial cells colonize the legume's root nodules, ______ dense clusters, which can enhance the plant's uptake of atmospheric nitrogen.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to produce
B) having produced
C) produce
D) producing

Question 16

Skill: Standard English Conventions (Punctuation/Appositives) | Difficulty: Hard

Though she would go on to collaborate with many distinguished choreographers, including Pina Bausch on her production *Café Müller*, German ______ may be best remembered for her years with the Stuttgart Ballet, where she helped redefine the role of expressive movement in contemporary dance.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) dancer, Marcia Haydée
B) dancer Marcia Haydée,
C) dancer Marcia Haydée
D) dancer, Marcia Haydée,

Question 17

Skill: Standard English Conventions (Punctuation) | Difficulty: Hard

Dr. Helena Johansson's monograph on "geothermal vents," formations through which superheated water escapes from the ocean floor, is included in the compendium *Abyssal Frontiers: New Perspectives in Deep-Sea Research*. The compendium was not written solely by ______ other researchers, such as Dr. Kenji Morimoto and Dr. Fatima Osei, also contributed sections.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Johansson, however,
B) Johansson; however,
C) Johansson. However,
D) Johansson, however;

Question 18

Skill: Standard English Conventions (Punctuation) | Difficulty: Medium

The World Health Organization (WHO) compiles comparative health expenditure data for its member states. According to this data, in 2020, a standard package of medical services costing 100 US dollars (USD) in the United States would have cost 58 USD in member ______

Which choice completes the text so that it conforms to the conventions of Standard English?

A) state Mexico.
B) state: Mexico.
C) state, Mexico.
D) state—Mexico.

Question 19

Skill: Transitions | Difficulty: Hard

In his mosaic *The Tree of Life*, Gustav Klimt endeavors to portray natural growth in four-dimensional space. Obviously, Klimt's artwork cannot literally render a fourth dimension; human vision is restricted to three spatial dimensions. ______ by depicting the tree's branches through spiraling, overlapping gold-leaf patterns that suggest movement through time, Klimt evokes a tantalizing sense of what such a higher-dimensional reality might feel like.

Which choice completes the text with the most logical transition?

A) For instance,
B) Even so,
C) To clarify,
D) Additionally,

Question 20

Skill: Transitions | Difficulty: Hard

In 2019, geophysicist Dr. Amara Osei recalculated the geographic center of the African continent, which had previously been established in 1951 by the Institut Géographique National. The Institut, Osei contended, had neglected important factors, including the curvature of the continental shelf. ______ the Institut's designated center—a point 80 kilometers northeast of Bangui—was inaccurate.

Which choice completes the text with the most logical transition?

A) Furthermore,
B) Alternatively,
C) Consequently,
D) Conversely,

Question 21

Skill: Transitions | Difficulty: Medium

On June 18, 1932, aviator Amelia Earhart was honored with a state dinner in Dublin, Ireland, in recognition of her solo transatlantic crossing. Of the 147 state dinners hosted between 1920 and 1950, a notable proportion celebrated achievements in aviation. _______ Earhart's dinner was one of just 9 honoring solo aviators.

Which choice completes the text with the most logical transition?

A) Additionally,

B) Hence,

C) In fact,

D) However,

Question 22

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• The Harlem Renaissance was a cultural and artistic movement centered in New York City during the 1920s and 1930s.
• Langston Hughes was a poet and novelist who became active in the movement in 1921.
• Zora Neale Hurston was an anthropologist and author who became active in the movement in 1925.
• Hughes is known for pioneering jazz poetry.
• Hurston is known for her ethnographic studies of African American folklore.

The student wants to emphasize a similarity between Hughes and Hurston. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Hughes pioneered jazz poetry, while Hurston focused on ethnographic studies of folklore.

B) Both Hughes and Hurston were active participants in the Harlem Renaissance, a cultural movement centered in New York City.

C) Hughes became active in the Harlem Renaissance in 1921, four years before Hurston.

D) Hurston, an anthropologist and author, is remembered for her ethnographic studies of African American folklore.

Question 23

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• Isotope dating is a technique for determining the age of geological specimens.

• Age can be expressed as a radiometric date.

• A radiometric date is calculated from the ratio of a radioactive isotope's remaining quantity to its original quantity in a specimen.

• The radiometric date of a zircon crystal from the Jack Hills formation in Western Australia is 4.374 billion years.

• The radiometric date of a feldspar grain from the Nuvvuagittuq greenstone belt in Quebec, Canada, is 4.280 billion years.

The student wants to specify the age of the Jack Hills zircon relative to that of all known mineral specimens. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The radiometric date of the Jack Hills zircon is 4.374 billion years.

B) The Jack Hills zircon's age relative to all known minerals is 4.280 billion years.

C) The age of the Jack Hills zircon can be expressed by a radiometric date, a measure derived from the ratio of remaining radioactive isotopes in the specimen.

D) The Jack Hills zircon is older than the Nuvvuagittuq feldspar, as reflected by the zircon's greater radiometric date.

Question 24

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

• Generally, a gas will compress when cooled.

• The compression of a gas due to cooling is described by Charles's Law.

• A 2021 study led by Dr. Rosa Ferreira and Dr. Akio Tanaka measured the thermal contraction of various noble gases.

• When a 500 mL sample of argon gas was cooled from 300 K to 250 K, its volume decreased by 83.3 mL.

• When a 500 mL sample of krypton gas was cooled from 300 K to 200 K, its volume decreased by 166.7 mL.

The student wants to contrast the two gas samples. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The krypton sample in the 2021 study was cooled through a larger temperature range than the argon sample and consequently contracted more.

B) Cooling a gas will typically cause it to decrease in volume, a relationship described by Charles's Law.

C) When the gases were cooled as part of the 2021 study, the volume of both samples shrank.

D) In 2021, researchers measured the thermal contraction of various noble gases, including samples of argon and krypton.

Question 25

Skill: Command of Evidence (Textual) | Difficulty: Hard

Prospect theory, developed by psychologists Daniel Kahneman and Amos Tversky, predicts that people weight potential losses more heavily than equivalent potential gains—a phenomenon termed loss aversion. Dr. Mei-Ling Chen and colleagues designed an experiment in which participants were offered a gamble: a 50% chance to win $150 or a 50% chance to lose $100. Despite the positive expected value, a majority of participants declined the gamble. Chen and colleagues concluded that loss aversion, rather than rational expected-value calculation, governed the participants' decisions.

Which finding, if true, would most directly support Chen and colleagues' conclusion?

A) When the potential gain was increased to $300 while the potential loss remained $100, most participants still declined the gamble.

B) Participants who had recently experienced a financial windfall were more likely to accept the gamble than those who had not.

C) When the gamble was reframed as avoiding a $100 penalty (rather than risking a $100 loss), acceptance rates increased significantly.

D) Participants with advanced training in statistics accepted the gamble at rates similar to those of participants without such training.

Question 26

Skill: Command of Evidence (Quantitative) | Difficulty: Hard

[PLACEHOLDER: Table titled 'Studies of Songbird Population Density in Temperate Forests' with columns: Research site | Location | Method | Min density (birds/km²) | Max density (birds/km²) | Density range. Rows: Site A | Oregon (US) | ground transects | 1.80 | 4.10 | 2.30 / Site B | Bavaria (Germany) | camera traps | 2.45 | 5.90 | 3.45 / Site C | Hokkaido (Japan) | drone thermal imaging | 3.60 | 8.80 | 5.20 / Site D | British Columbia (Canada) | acoustic monitoring | 1.15 | 1.50 | 0.35. Below: 'A researcher hypothesizes that acoustic monitoring produces the narrowest density estimate range among all survey methods tested, suggesting it is the most precise technique.']

Which choice best describes data from the table that challenge the researcher's hypothesis?

A) Site C, which used drone-based thermal imaging, produced a density estimate range of 5.20, the largest among all methods.

B) Site D, which used acoustic monitoring, produced a narrower density estimate range than Site A, which used ground transects.

C) Site B, which used camera traps, produced a density estimate range larger than that of Site A, which used ground transects.

D) Site A, which used ground transects, produced a minimum density estimate of 1.80 individuals per square kilometer.

Question 27

Skill: Inferences | Difficulty: Hard

The Sapir-Whorf hypothesis, in its strong form, posits that the structure of a language determines the cognitive patterns of its speakers. Dr. Tomás Guerrero and colleagues tested this by examining spatial reasoning in speakers of Tzeltal (a Mayan language that uses absolute directional terms rather than relative ones like "left" and "right") and speakers of Dutch. The researchers found that Tzeltal speakers solved certain spatial rotation tasks faster than Dutch speakers but performed comparably on tasks involving object recognition. Guerrero and colleagues therefore concluded that ______

Which choice most logically completes the text?

A) linguistic structure influences certain categories of spatial cognition without uniformly determining all aspects of thought.

B) the strong form of the Sapir-Whorf hypothesis is fully supported by the experimental evidence.

C) Tzeltal speakers possess generally superior cognitive abilities compared to Dutch speakers.

D) the differences in task performance were attributable to cultural familiarity with the test materials rather than to linguistic structure.

Section 2: Math

Module 1 (22 Questions — All Students)

Question 1

Skill: Linear Equations in Two Variables (Graph) | Difficulty: Easy

[PLACEHOLDER: Coordinate plane showing two lines intersecting at the point (3, -2). One line has a positive slope passing through approximately (-3, -6) and (3, -2). The other line has a negative slope passing through approximately (-1, 2) and (3, -2). Axes range from -6 to 6 on both x and y.]

The graph of a system of two linear equations is shown. The solution to the system is (x, y). What is the value of y?

A) -3
B) -2
C) 4
D) 5

Question 2

Skill: Function Evaluation | Difficulty: Easy

The function g is defined by $g(x) = 12x + 7$. What is the value of $g(x)$ when $x = 3$?

A) 29
B) 36
C) 43
D) 57

Question 3

Skill: Linear Equations (Slope-Intercept) | Difficulty: Easy

In the xy-plane, the graph of $y = f(x)$ is a line that passes through (0, 17) and has a slope of 5. Which equation defines f?

A) $f(x) = 17x$
B) $f(x) = 17x + 5$
C) $f(x) = 5x$
D) $f(x) = 5x + 17$

Question 4

Skill: Linear Relationships (Tables) | Difficulty: Easy

[PLACEHOLDER: Table with columns x and y. Rows: x=0, y=30 | x=1, y=31 | x=2, y=32]

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A) $y = x + 30$

B) $y = 32x$

C) $y = 32x + 2$

D) $y = 30x$

Question 5

Skill: Systems of Equations (Infinitely Many Solutions) | Difficulty: Medium

$y = (3/7)x + 4$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b?

A) -4

B) -3/7

C) 3/7

D) 4

Question 6

Skill: Percentage and Proportion | Difficulty: Easy

A store sells notebooks for $4 each. During a sale, the price of each notebook is reduced by 25%. What is the sale price, in dollars, of each notebook?

A) $1.00

B) $2.00

C) $3.00

D) $3.25

Question 7

Skill: Exponential Functions | Difficulty: Medium

The relationship between the variables x and y is defined by an exponential equation. When $x = 0$, the value of y is 75, and for every increase in the value of x by 1, the corresponding value of y increases by 15% of its previous value. Which equation represents this relationship?

A) $y = 75(1.15)^x$

B) $y = 75(1.015)^x$

C) $y = 15(1.75)^x$

D) $y = 15(1.075)^x$

Question 8

Skill: Statistics (Mean) | Difficulty: Easy

[PLACEHOLDER: Table with columns: Session | Laps. Rows: A: 10 | B: 8 | C: 16 | D: 14 | E: 12]

The table shows the number of laps a swimmer completed in each of 5 practice sessions. What was the mean number of laps per session for this swimmer?

A) 10
B) 12
C) 14
D) 16

Question 9

Skill: Absolute Value Functions | Difficulty: Easy

The function h is defined by $h(x) = 9|x|$. What is the value of $h(-3)$?

A) -27
B) 6
C) 12
D) 27

Question 10

Skill: Line of Best Fit (Slope) | Difficulty: Medium

[PLACEHOLDER: Scatterplot with line of best fit. X-axis from 0 to 8, Y-axis from 0 to 18. Data points approximately at (0, 14), (1, 13), (2, 11.5), (3, 10), (4, 9), (5, 8), (6, 7), (7, 5.5). Line of best fit goes from approximately (0, 14) to (8, 5), showing negative slope of about -1.12.]

Which of the following is closest to the slope of the line of best fit shown?

A) -9.42
B) -1.12
C) 1.12
D) 9.42

Question 11

Skill: Geometry (Parallel Lines and Transversals) | Difficulty: Medium

[PLACEHOLDER: Figure showing two parallel horizontal lines (p and q) cut by a transversal line (t). The angle between the transversal and line p is labeled 80°. The corresponding angle on the other side of the transversal at line q is labeled y°. Note: Figure not drawn to scale.]

In the figure, line p is parallel to line q, and both lines are intersected by line t. If $y = 3x + 5$, what is the value of x?

A) 25
B) 35
C) 43.75
D) 55

Question 12

Skill: Linear Equations (Word Problems) | Difficulty: Easy

Priya purchased 7 identical scarves at the same price per scarf. She applied a discount voucher worth $35 to the total. After using the voucher, the cost for all the scarves was $91. What was the original price, in dollars, of 1 scarf?

(Student-produced response — grid-in)

Question 13

Skill: Linear Equations (Point-Slope) | Difficulty: Medium

Line *m* in the *xy*-plane has a slope of -1/4 and passes through the point (8, 5). Which equation defines line *m*?

A) $y = 8x - 1/4$

B) $y = 5x + 8$

C) $y = -x/4 + 7$

D) $y = -x/4 + 5$

Question 14

Skill: Solving Equations (Factored Form) | Difficulty: Easy

$(x - 14)(x - 9)(x + 5)(x + 11) = 0$

What is a positive solution to the given equation?

(Student-produced response — grid-in)

Question 15

Skill: Geometry (Volume of a Cone) | Difficulty: Medium

A right circular cone has a height of 15 centimeters (cm) and a base with a radius of 4 cm. What is the volume, in cm^3, of this cone?

A) 20π

B) 60π

C) 80π

D) 240π

Question 16

Skill: Linear Models | Difficulty: Medium

A linear model estimates the enrollment at a university from 2000 to 2024. The model estimates the enrollment was 8,000 students in 2000, 14,500 students in 2012, and *x* students in 2024. To the nearest whole number, what is the value of *x*?

(Student-produced response — grid-in)

Question 17

Skill: Exponential Functions (Interpretation) | Difficulty: Medium

$y = 400(3)^{2x}$

The equation gives the estimated number of podcast subscribers at the end of every three-month period, x, starting from the end of March 2020, where $0 \leq x \leq 5$. Which statement is the best interpretation of the y-intercept of the graph of this equation in the xy-plane?

A) The estimated number of subscribers at the end of the first three-month period was 400.

B) The estimated number of subscribers at the end of March 2020 was 400.

C) The estimated number of subscribers at the end of March 2021 was 1,200.

D) The estimated number of subscribers at the end of March 2020 was 1,200.

Question 18

Skill: Systems of Equations | Difficulty: Hard

$x/5 + 3(y - 8) = 47$
$x/4 - 3(y - 8) = 23$

The solution to the given system of equations is (x, y). What is the value of $20x$?

(Student-produced response — grid-in)

Question 19

Skill: Inequalities (Word Problems) | Difficulty: Hard

When several capacitors are connected in parallel in a circuit, the total capacitance is the sum of the individual capacitances. A certain circuit consists of 10 capacitors connected in parallel, where the capacitance of each capacitor is positive. This circuit has a total capacitance of 200 microfarads. The combined capacitance of 4 of the capacitors is 120 microfarads. Which inequality best represents all possible values of the capacitance x, in microfarads, of one of the other 6 capacitors in this circuit?

A) $0 < x < 10$

B) $0 < x < 80$

C) $80 < x < 120$

D) $80 < x < 200$

Question 20

Skill: Linear Functions (Interpretation) | Difficulty: Medium

A library charges a membership fee and then a fixed amount for each book borrowed. The function $f(n) = 25 - 2n$ gives the remaining balance, in dollars, on a member's account after borrowing n books. Which of the following represents the amount, in dollars, deducted from the account each time a book is borrowed?

A) $2n$

B) 2

C) 25

D) $25 - 2n$

Question 21

Skill: Solving Equations | Difficulty: Medium

If $7 - 3(5 - 2x) = 4 - 6(5 - 2x)$, what is the value of $5 - 2x$?

(Student-produced response — grid-in)

Question 22

Skill: Percentages | Difficulty: Medium

On a ranch, 48.0% of the acreage is grazing land and the remaining acreage is cropland. There are barns on exactly 18.5% of the grazing land acreage, and there are barns on exactly 12.0% of the cropland acreage. If there are barns on exactly p% of the total acreage of the ranch, what is the value of p?

(Student-produced response — grid-in)

Module 2 — Easier Path (22 Questions)

Approximately 70% Easy/Medium, 30% Hard

Question 1

Skill: Geometry (Triangle Angles) | Difficulty: Easy

[PLACEHOLDER: Right triangle with one angle marked 90° (right angle at the bottom-right vertex), another angle marked 28° (at the top vertex), and the third angle marked a° (at the bottom-left vertex). Note: Figure not drawn to scale.]

In the right triangle shown, what is the value of a?

A) 18
B) 62
C) 72
D) 90

Question 2

Skill: Function Evaluation (Graph) | Difficulty: Easy

The graph of the polynomial function f in the xy-plane, where $y = f(x)$, passes through the point (7, 2). What is the value of $f(7)$?

A) 2
B) 7
C) 9
D) 14

Question 3

Skill: Geometry (Circles) | Difficulty: Easy

A circle has a circumference of 56π centimeters. What is the diameter, in centimeters, of the circle?

(Student-produced response — grid-in)

Question 4

Skill: Rearranging Formulas | Difficulty: Medium

$36 = 2K / v^2$

For an object with a mass of 36 kilograms, the given equation relates the kinetic energy K, in joules, to the object's speed v, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed in terms of its kinetic energy?

A) $v = \sqrt{(2K/36)}$
B) $v = \sqrt{(36/2K)}$
C) $v = 36^2/2K$
D) $v = 2K/36^2$

Question 5

Skill: Function Evaluation (Quadratic) | Difficulty: Easy

The function f is defined by $f(x) = 3x^2$. What is the value of $f(10)$?

A) 30
B) 60
C) 100
D) 300

Question 6

Skill: Exponential Functions (Y-intercept) | Difficulty: Medium

$g(x) = (1/7)(4)^x$

If the given function g is graphed in the xy-plane, where $y = g(x)$, what is the y-intercept of the graph?

A) (0, 0)
B) (0, 1/7)
C) (0, 4/7)
D) (0, 4)

Question 7

Skill: Ratios and Proportions | Difficulty: Easy

A recipe calls for 3 cups of flour for every 2 cups of sugar. If a baker uses 9 cups of flour, how many cups of sugar should the baker use?

A) 4
B) 5
C) 6
D) 7

Question 8

Skill: Exponential Growth/Decay | Difficulty: Medium

The population of a certain town tripled every 60 years from 1700 to 1940. The population of this town was 81,000 in 1940. What was the population of this town in 1700?

(Student-produced response — grid-in)

Question 9

Skill: Right Triangles (Trigonometry) | Difficulty: Medium

[PLACEHOLDER: Right triangle PQR with the right angle at Q. The hypotenuse PR is labeled 60. Vertices: P at top, Q at bottom-right (right angle), R at bottom-left. Note: Figure not drawn to scale.]

In right triangle *PQR*, the tangent of angle *R* is 3/5. What is the length of *PQ*?

A) 24/5
B) 12/5
C) 36
D) 60

Question 10

Skill: Linear Equations (Word Problems) | Difficulty: Medium

A streaming service charges $8 for the first movie rented during the month and an additional $5 for each movie after the first. Which equation describes this situation, where y is the total cost, in dollars, for movies rented during a month, x is the number of movies rented during the month, and $x > 0$?

A) $y = x + 8$
B) $y = 5x + 8$
C) $y = 5(x - 1) + 8$
D) $y = (x - 1) + 8$

Question 11

Skill: Algebraic Equations (Infinite Solutions) | Difficulty: Hard

$$a(5x - 10) + 3a = 5(ax - 2a) - 21$$

In the given equation, a is a constant. The equation has infinitely many solutions. What are all possible values of a?

A) 0 only
B) -3 only
C) Any real number
D) No real number

Question 12

Skill: Percentages (Algebraic) | Difficulty: Medium

If $x > 0$ and x is 25% of y, which expression represents y in terms of x?

A) $(100/25)x$
B) $(25/100)x$
C) $(1/25)x$
D) $(75/100)x$

Question 13

Skill: Exponent Rules | Difficulty: Medium

■(x^m)

In the given expression, m is a constant and $x > 1$. The expression can be rewritten as ■x. What is the value of m?

(Student-produced response — grid-in)

Question 14

Skill: Exponential Decay (Percent Decrease) | Difficulty: Hard

$z(w) = (0.743)^{2w}$

The function z is defined by the equation shown. The value of $z(w)$ decreases by p% for each increase by 1 in the value of w. Which of the following is closest to the value of p?

A) 0.448
B) 0.552
C) 44.8
D) 55.2

Question 15

Skill: Linear Functions (Graph Interpretation) | Difficulty: Hard

[PLACEHOLDER: Graph showing a line with positive y-intercept (approximately y = 6 when x = 0) and a steep negative slope, crossing the x-axis at approximately x = 3. The line goes from upper-left to lower-right. Axes: x from -4 to 5, y from -2 to 10.]

The graph of the linear function $y = f(x) + 9$ is shown. If c and d are positive constants, which equation could define f?

A) $f(x) = -d - cx$
B) $f(x) = d + cx$
C) $f(x) = d - cx$
D) $f(x) = -d + cx$

Question 16

Skill: Unit Conversion (Rates) | Difficulty: Medium

On average, a certain factory produces 63 widgets every m hours. At this rate, which expression represents the number of widgets, on average, the factory produces every k days?

A) $63m / (24k)$
B) $63k / (24m)$
C) $(24 \times 63 \times m) / k$
D) $(24 \times 63 \times k) / m$

Question 17

Skill: Quadratic Equations (Product of Solutions) | Difficulty: Hard

$15x^2 + (15s + r)x + rs = 0$

In the given equation, *r* and *s* are positive constants. The product of the solutions to the given equation is *krs*, where *k* is a constant. What is the value of *k*?

(Student-produced response — grid-in)

Question 18

Skill: Percentages (Speed/Rate) | Difficulty: Hard

The velocities of particles X, Y, and Z are *a* meters per second, *b* meters per second, and *c* meters per second, respectively. If the velocity of particle X is 5,000% of the velocity of particle Z and the velocity of particle Z is 0.006% of the velocity of particle Y, which expression represents the value of *a* + *b* in terms of *c*?

A) 16,717*c*
B) 5,006*c*
C) 5,000.06*c*
D) 1,717*c*

Question 19

Skill: Polynomial Expansion | Difficulty: Hard

$-9(3x - 4)^2 + 5(3x - 1)^2$

The given expression can be rewritten as $(a/4)x^2 + (b/4)x + (c/4)$, where *a*, *b*, and *c* are constants. What is the value of *a* + *b* + *c*?

A) -128
B) -64
C) -36
D) -9

Question 20

Skill: Exponential Functions (Percent Change) | Difficulty: Hard

$f(x) = 52(b)^x$

In the given function, *b* is a positive constant. For every increase in the value of *x* by 1, the value of *f*(*x*) increases by *c*%, where 0 < *c* < 100. Which expression gives the value of *c* in terms of *b*?

A) 1 + *b*/100
B) *b* + 100
C) 100(*b* + 1)
D) 100(*b* - 1)

Question 21

Skill: Unit Conversion (Rates) | Difficulty: Medium

On average, a particular vine grows 52 centimeters every m weeks. At this rate, which expression represents the number of centimeters, on average, the vine grows every k months (assume 1 month ≈ 4 weeks)?

A) $52m / (4k)$

B) $52k / (4m)$

C) $(4 \times 52 \times m) / k$

D) $(4 \times 52 \times k) / m$

Question 22

Skill: Geometry (Circles and Chords) | Difficulty: Hard

[PLACEHOLDER: Circle with points P, Q, R, T on the circumference. Chord PR and chord QT intersect at interior point S at right angles. PQ < QR. QS = √410. Diameter = 205. Note: Figure not drawn to scale.]

In the figure shown, points P, Q, R, and T lie on the circle, and $PQ < QR$. Segment PR is perpendicular to segment QT at point S, and $QS = \sqrt{410}$. The diameter of the circle is 205. If $RS/PS = r$, what is the value of r?

(Student-produced response — grid-in)

Module 2 — Harder Path (22 Questions)

Approximately 30% Easy/Medium, 70% Hard

Question 1

Skill: Geometry (Triangle Angles) | Difficulty: Easy

[PLACEHOLDER: Right triangle with one angle marked 90° (right angle at the bottom-left vertex), another angle marked 19° (at the bottom-right vertex), and the third angle marked b° (at the top vertex). Note: Figure not drawn to scale.]

In the right triangle shown, what is the value of *b*?

A) 19
B) 71
C) 90
D) 109

Question 2

Skill: Function Evaluation | Difficulty: Easy

The graph of the polynomial function *f* in the *xy*-plane, where *y* = *f*(*x*), passes through the point (4, 6). What is the value of *f*(4)?

A) 4
B) 6
C) 10
D) 24

Question 3

Skill: Geometry (Circles) | Difficulty: Easy

A circle has a circumference of 38π centimeters. What is the diameter, in centimeters, of the circle?

(Student-produced response — grid-in)

Question 4

Skill: Rearranging Formulas | Difficulty: Medium

$64 = 2K / v^2$

For an object with a mass of 64 kilograms, the given equation relates the kinetic energy *K*, in joules, to the object's speed *v*, in meters per second, where *K* and *v* are positive. Which equation correctly expresses this object's speed in terms of its kinetic energy?

A) $v = \sqrt{(2K/64)}$
B) $v = \sqrt{(64/2K)}$
C) $v = 64^2/2K$
D) $v = 2K/64^2$

Question 5

Skill: Function Evaluation (Quadratic) | Difficulty: Medium

The function f is defined by $f(x) = 5x^2$. What is the value of $f(9)$?

A) 45
B) 90
C) 225
D) 405

Question 6

Skill: Exponential Functions (Y-intercept) | Difficulty: Medium

$g(x) = (1/11)(7)^x$

If the given function g is graphed in the xy-plane, where $y = g(x)$, what is the y-intercept of the graph?

A) (0, 0)
B) (0, 1/11)
C) (0, 7/11)
D) (0, 7)

Question 7

Skill: Systems of Equations (Complex) | Difficulty: Hard

$x/6 + 4(y - 3) = 52$
$x/2 - 4(y - 3) = 18$

The solution to the given system of equations is (x, y). What is the value of $12x$?

(Student-produced response — grid-in)

Question 8

Skill: Exponential Growth | Difficulty: Hard

The population of a certain colony of bacteria quadrupled every 50 minutes from the start of an experiment to 150 minutes later. The population was 6,400 at 150 minutes. What was the population at the start of the experiment?

(Student-produced response — grid-in)

Question 9

Skill: Right Triangles (Trigonometry) | Difficulty: Hard

[PLACEHOLDER: Right triangle DEF with the right angle at E. The hypotenuse DF is labeled 84. Vertices: D at top, E at bottom-right (right angle), F at bottom-left. Note: Figure not drawn to scale.]

In right triangle *DEF*, the tangent of angle *F* is 5/7. What is the length of *DE*?

A) 84/5

B) 12/7

C) 60

D) 84

Question 10

Skill: Algebraic Equations (Infinite Solutions) | Difficulty: Hard

$a(9x - 18) + 7a = 9(ax - 2a) - 63$

In the given equation, *a* is a constant. The equation has infinitely many solutions. What are all possible values of *a*?

A) 0 only

B) -9 only

C) Any real number

D) No real number

Question 11

Skill: Percentages (Algebraic) | Difficulty: Medium

If $x > 0$ and *x* is 40% of *y*, which expression represents *y* in terms of *x*?

A) (100/40)x

B) (40/100)x

C) (1/40)x

D) (60/100)x

Question 12

Skill: Exponent Rules | Difficulty: Hard

■√(x^m)

In the given expression, *m* is a constant and $x > 1$. The expression can be rewritten as ■*x*. What is the value of *m*?

(Student-produced response — grid-in)

Question 13

Skill: Linear Functions (Word Problems) | Difficulty: Medium

A music subscription service charges $15 for the first album downloaded in a month and an additional $9 for each album after the first. Which equation describes this situation, where y is the total cost in dollars, x is the number of albums downloaded during the month, and $x > 0$?

A) $y = x + 15$

B) $y = 9x + 15$

C) $y = 9(x - 1) + 15$

D) $y = (x - 1) + 15$

Question 14

Skill: Exponential Decay (Percent Decrease) | Difficulty: Hard

$$z(w) = (0.917)^{2w}$$

The function z is defined by the equation shown. The value of $z(w)$ decreases by p% for each increase by 1 in the value of w. Which of the following is closest to the value of p?

A) 0.159

B) 0.841

C) 15.9

D) 84.1

Question 15

Skill: Linear Functions (Graph Interpretation) | Difficulty: Hard

[PLACEHOLDER: Graph showing a line with a positive y-intercept (approximately y = 4 when x = 0) and a steep negative slope, crossing the x-axis near x = 2. Line goes from upper-left to lower-right. Axes: x from -5 to 5, y from -4 to 12.]

The graph of the linear function $y = f(x) + 7$ is shown. If c and d are positive constants, which equation could define f?

A) $f(x) = -d - cx$

B) $f(x) = d + cx$

C) $f(x) = d - cx$

D) $f(x) = -d + cx$

Question 16

Skill: Exponential Functions (Percent Change) | Difficulty: Hard

$f(x) = 84(b)^x$

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by c%, where $0 < c < 100$. Which expression gives the value of c in terms of b?

A) $1 + b/100$

B) $b + 100$

C) $100(b + 1)$

D) $100(b - 1)$

Question 17

Skill: Quadratic Equations (Product of Solutions) | Difficulty: Hard

$19x^2 + (19s + r)x + rs = 0$

In the given equation, r and s are positive constants. The product of the solutions to the given equation is krs, where k is a constant. What is the value of k?

(Student-produced response — grid-in)

Question 18

Skill: Percentages (Speed/Rate) | Difficulty: Hard

The velocities of objects M, N, and P are *a* meters per second, *b* meters per second, and *c* meters per second, respectively. If the velocity of object M is 4,000% of the velocity of object P and the velocity of object P is 0.005% of the velocity of object N, which expression represents the value of $a + b$ in terms of c?

A) $20{,}040c$

B) $4{,}020c$

C) $4{,}000.05c$

D) $24{,}000c$

Question 19

Skill: Polynomial Expansion | Difficulty: Hard

$-12(4x - 5)^2 + 3(4x - 3)^2$

The given expression can be rewritten as $(a/4)x^2 + (b/4)x + (c/4)$, where a, b, and c are constants. What is the value of $a + b + c$?

A) -180

B) -108

C) -72

D) -27

Question 20

Skill: Unit Conversion (Rates) | Difficulty: Hard

On average, a certain printer produces 48 pages every m minutes. At this rate, which expression represents the number of pages, on average, the printer produces every k hours?

A) $48m / (60k)$
B) $48k / (60m)$
C) $(60 \times 48 \times m) / k$
D) $(60 \times 48 \times k) / m$

Question 21

Skill: Inequalities (Word Problems) | Difficulty: Hard

When several light bulbs are connected in series in a circuit, the total wattage is the sum of the individual wattages. A certain strand consists of 12 light bulbs in series, where each bulb's wattage is positive. The total wattage of the strand is 300 watts. The combined wattage of 5 of the bulbs is 180 watts. Which inequality best represents all possible values of the wattage w of one of the other 7 bulbs?

A) $0 < w < 12$
B) $0 < w < 120$
C) $120 < w < 180$
D) $120 < w < 300$

Question 22

Skill: Geometry (Circles and Chords) | Difficulty: Hard

[PLACEHOLDER: Circle with points W, X, Y, Z on the circumference. Chord WY and chord XZ intersect at interior point V at right angles. WX < XY. XV = √520. Diameter = 260. Note: Figure not drawn to scale.]

In the figure shown, points W, X, Y, and Z lie on the circle, and $WX < XY$. Segment WY is perpendicular to segment XZ at point V, and $XV = \sqrt{520}$. The diameter of the circle is 260. If $YV/WV = r$, what is the value of r?

(Student-produced response — grid-in)

Question Count Verification

Section	Module	Count
S1 Reading & Writing	Module 1	27
S1 Reading & Writing	Module 2 Easier	27
S1 Reading & Writing	Module 2 Harder	27
S2 Math	Module 1	22
S2 Math	Module 2 Easier	22
S2 Math	Module 2 Harder	22
	TOTAL	**147**

Note: The original uploaded test contained one Module 2 path per section. This practice test adds both Easier and Harder Module 2 paths for full adaptive coverage. Total: 147 questions (S1: 81, S2: 66).